Introduction to Supply Chain Management

Course on

Supply Chain Management

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Purchasing options ?

How did it reach to you?

What was the manufacturing process?

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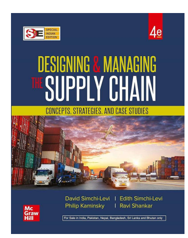
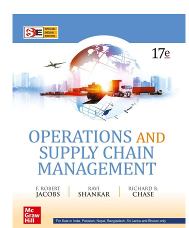
Reference Book for this course





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Where it all began…

Henry Ford, 1913: First Moving Assembly Line

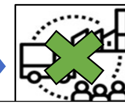
Manufacturing process designed for sequential arrangement of workers, machines, tools and parts in such a away that motion of all these at minimum extent possible

- ✓ Manufacturing time drastically reduced
- ✓ Mass Production Boomed

WWII, 1945: Mass Production

After WWII mass production prioritized to reduce costs by economies of scale as strategy.

- ✓ Mass Production Need
- ✓ Economies of Scale



1950-60s: SCM Unknown

SCM was not prioritized.

- ✓ NPD took time
- ✓ Logistics costs high
- ✓ Poor quality

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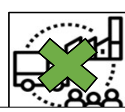
Where it all began…

Henry Ford, 1913: First Moving Assembly Line

WWII, 1945: Mass Production



1950-60s: SCM Unknown

- ✓ Awareness of MRP.
- ✓ Focus on Increased performance
- ✓ MRPII – MRPI + Finances/General accounting + Demand Forecasting + Labour Planning + Capacity
- ✓ Strategic for Manufacturers

1970-80s: MRPI & MRPII

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Where it all began…

Henry Ford, 1913: First Moving Assembly Line

WWII, 1945: Mass Production

1950-60s: SCM Unknown

Manufacturers dealing with increased demands outsourced logistics to focus on Core Competencies

- ✓ Economic service, efficiency, productivity
- ✓ Relationship oriented approach: Customers & Suppliers
- ✓ TQM & JIT

1980-90s: Outsourcing & Core Competencies

1970-80s: MRPI & MRPII

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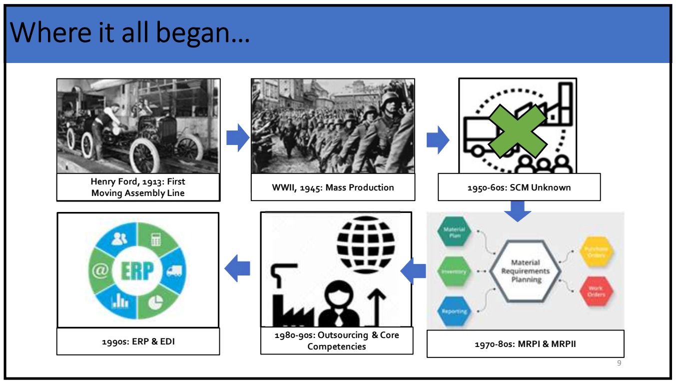
Where it all began…
Henry Ford, 1913: First Moving Assembly Line
WWII, 1945: Mass Production
1950-60s: SCM Unknown
1990s: ERP & EDI
ERP
1980-90s: Outsourcing & Core Competencies
Material Requirements Planning
1970-80s: MRPI & MRPII

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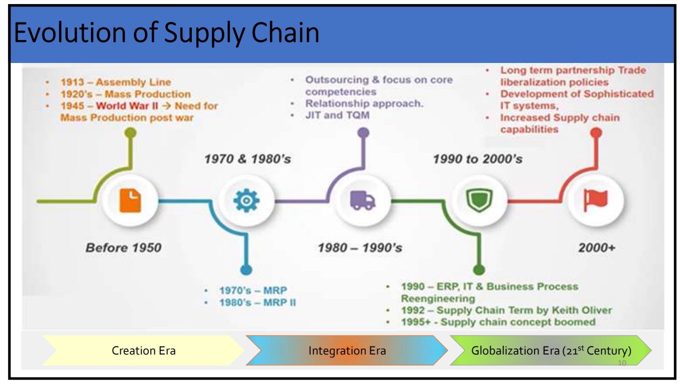
Evolution of Supply Chain
1913 – Assembly Line
1920's – Mass Production
1945 – World War II → Need for Mass Production post war
Outsourcing & focus on core competencies
Relationship approach.
JIT and TQM
Long term partnership Trade liberalization policies
Development of Sophisticated IT systems,
Increased Supply chain capabilities
1970 & 1980's
1990 to 2000's
Before 1950
1980 – 1990's
2000+
1970's – MRP
1980's – MRP II
1990 – ERP, IT & Business Process Reengineering
1992 – Supply Chain Term by Keith Oliver
1995+ - Supply chain concept boomed
Creation Era
Integration Era
Globalization Era (21st Century)

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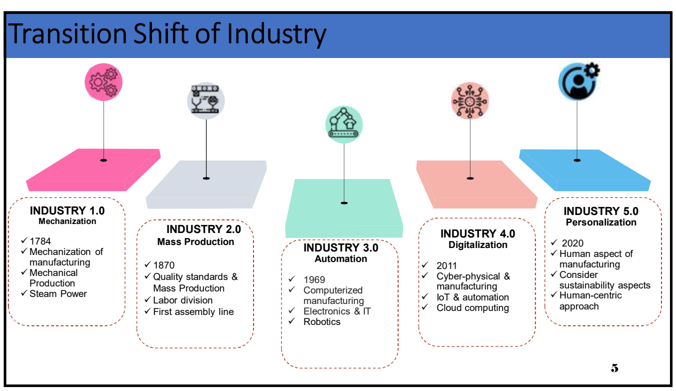
Transition Shift of Industry
INDUSTRY 1.0
Mechanization
✓1784
✓Mechanization of manufacturing
✓Mechanical Production
✓Steam Power
INDUSTRY 2.0
Mass Production
✓1870
✓Quality standards & Mass Production
✓Labor division
✓First assembly line
INDUSTRY 3.0
Automation
✓ 1969
✓ Computerized manufacturing
✓ Electronics & IT
✓ Robotics
INDUSTRY 4.0
Digitalization
✓ 2011
✓ Cyber-physical & manufacturing
✓ IoT & automation
✓ Cloud computing
INDUSTRY 5.0
Personalization
✓ 2020
✓Human aspect of manufacturing
✓Consider sustainability aspects
✓Human-centric approach
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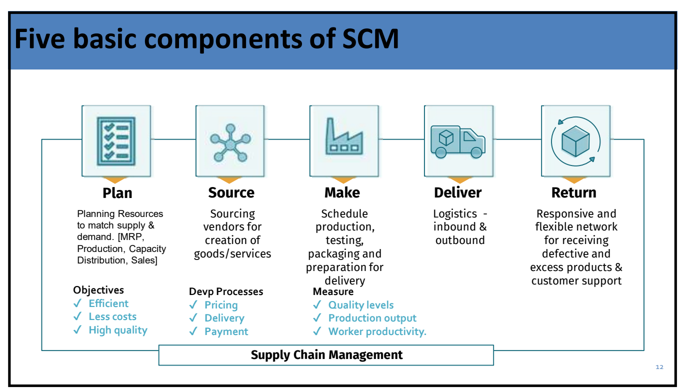
Five basic components of SCM
Plan
Planning Resources to match supply & demand. [MRP, Production, Capacity Distribution, Sales]
Source
Sourcing vendors for creation of goods/services
Make
Schedule production, testing, packaging and preparation for delivery
Deliver
Logistics - inbound & outbound
Return
Responsive and flexible network for receiving defective and excess products & customer support
Objectives
✓ Efficient
✓ Less costs
✓ High quality
Devp Processes
✓ Pricing
✓ Delivery
✓ Payment
Measure
✓ Quality levels
✓ Production output
✓ Worker productivity.
Supply Chain Management

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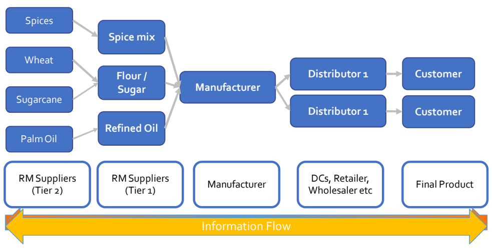
What is Supply Chain Management?

The activities required by an organization to deliver goods or services to the consumer. All processes from RM to final product.



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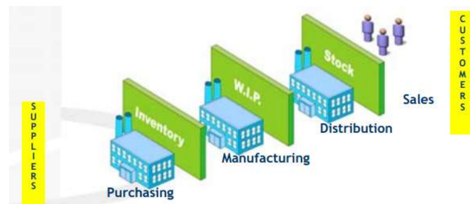
Evolution of Supply Chain- Stakeholders



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Evolution of Supply Chain- Stakeholders

DEFINITION OF SUPPLY CHAIN

FLOW OF GOODS & INFORMATION

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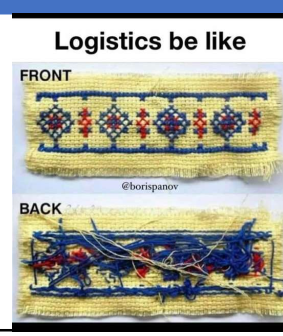
Evolution of Supply Chain



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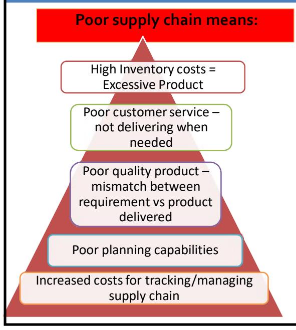
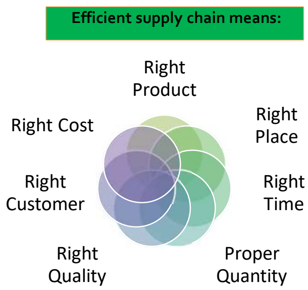
Why Supply Chain Management?





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Objectives of SCM

1. To maximize the overall value generated
2. To achieve maximum supply chain profitability
3. To reduce the supply chain costs to the minimum possible level
4. To generate synergy & transparency among the supply chain components
5. Upgrading Technology
6. To enhance company's top and bottom through improved manufacturing and service capability.
7. Faster response to market and building healthy customer- supplier responsiveness

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Drivers of SCM

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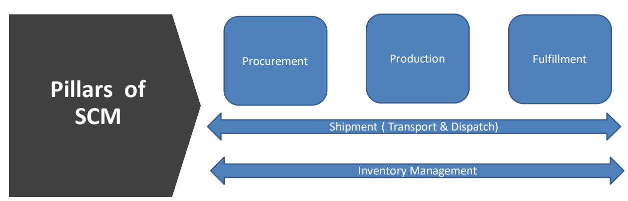

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Pillars of SCM

		Supply Chain Verticals		
Processes ->		Procurement	Production	Fullfillment
		Order Planning	Production Planning for FG	Order management
			Production schedule for FG	Lot sizing/ Bulk breaking
			Work in progress for intermidiate products (production lifecycle trace)	Fulfilment plan
Supply Chain Horizontals	Shipment	Transport planning	Transport planning	Transportation planning
		Dispatch Planning	Dispatch Planning	Dispatch planning
	Inventory	Inventory Management	Inventory management for raw materials and finished goods	Site-wise Replenishment

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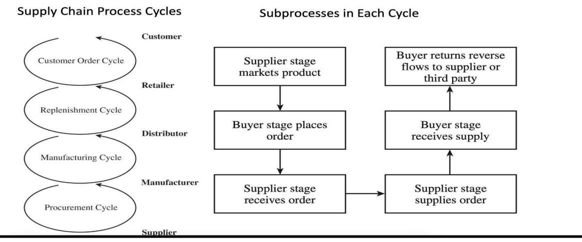
Cycle View of SCM

Cycle View of Supply Chain Processes

It specifies the roles and responsibilities of each member of the supply chain and the desired outcome for each process.



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Scope of Supply Chain Management

It Refers to range of **Activities**, **Processes** and **Functions** that are encompassed within the domain of Supply Chain Management

What is included in overall scope of managing a supply chain

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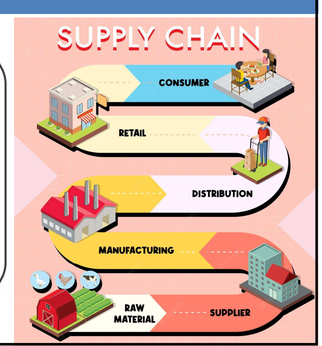
Scope of Supply Chain

Scope of Supply Chain Typically includes-

- Procurement
- Production & Manufacturing
- Inventory Management
- Logistics & Transportation
- Distribution & Fulfilment
- Demand Forecasting
- Supplier Relationship Management
- Information Flow
- Sustainability & Ethical Practices
- Risk Management



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1. Procurement

Sourcing Raw material, components, goods and services from suppliers

It encompasses activities like-
1. Supplier selection
2. Negotiation
3. Contract Manufacturing

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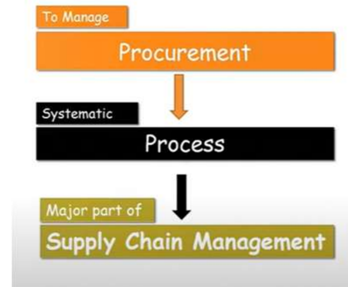
Role of Procurement in SCM



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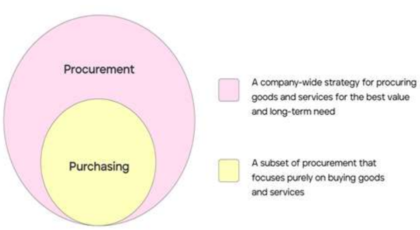
Procurement Versus Purchasing



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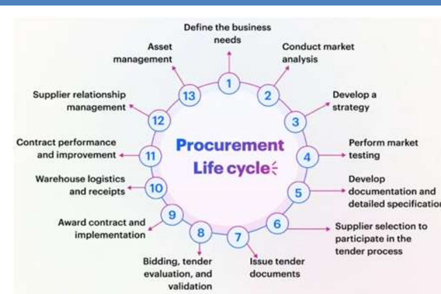
Procurement



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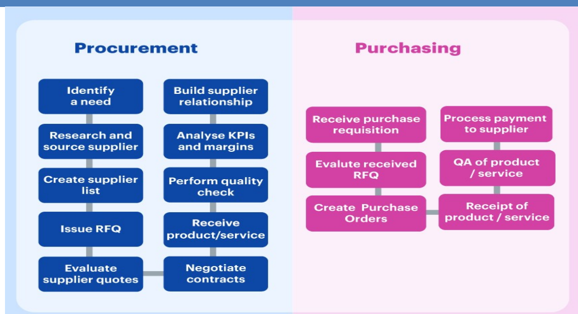
Procurement Versus Purchasing



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2. Production & Manufacturing

Managing the processes that transform the raw material & components into finished goods.

It includes-

1. Production Planning
2. Production Operation
3. Quality Control
4. Manufacturing efficiency

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What is Production Management

The set of **interrelated management activities which are involved in manufacturing** certain product is called Product Management.

Planning and control of industrial processes **to ensure that they move smoothly at the required level**.

Planning>> Organizing>> Directing>> Controlling the activities

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3. Inventory Management

How much inventory to keep?
Where to store it?
How to manage it efficiently?

- **Acquiring, keeping,** and **selling inventory,** including both raw materials and finished goods (products).
- It refers to having the **right stock**, at the **right levels**, in **the right place**, at **the right time,** and at the **right cost** and **price.**
- The practice to **guarantee** that there is **always enough stock to satisfy client orders** and **shortages are identified early**.
- **Inventory becomes revenue if** it is **sold**.
- Inventory **ties up cash before it sells**, even though it is reported as an **asset on the balance sheet**.

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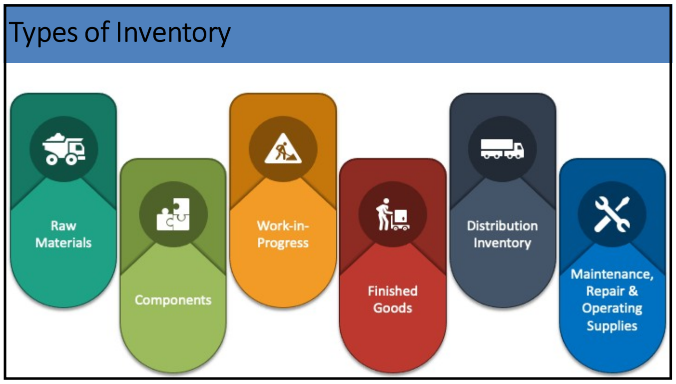
Types of Inventory
Raw Materials
Components
Work-in-Progress
Finished Goods
Distribution Inventory
Maintenance, Repair & Operating Supplies

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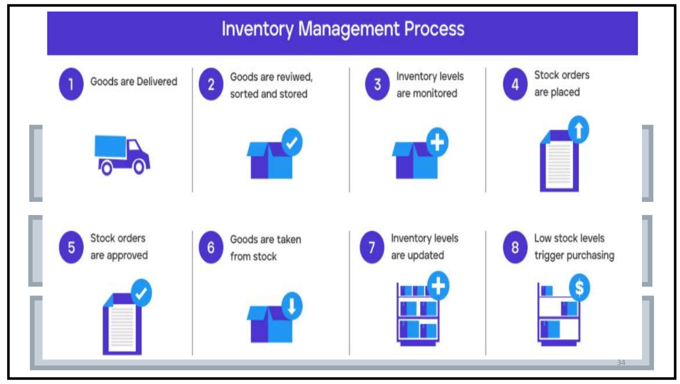
Inventory Management Process
1 Goods are Delivered
2 Goods are reviwed, sorted and stored
3 Inventory levels are monitored
4 Stock orders are placed
5 Stock orders are approved
6 Goods are taken from stock
7 Inventory levels are updated
8 Low stock levels trigger purchasing

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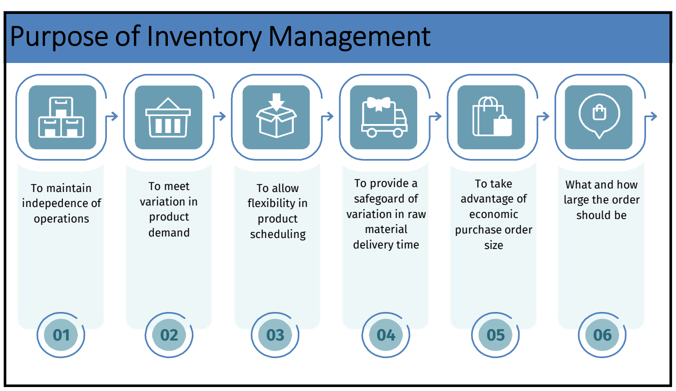
Purpose of Inventory Management
To maintain indepedence of operations
To meet variation in product demand
To allow flexibility in product scheduling
To provide a safegoard of variation in raw material delivery time
To take advantage of economic purchase order size
What and how large the order should be
01
02
03
04
05
06

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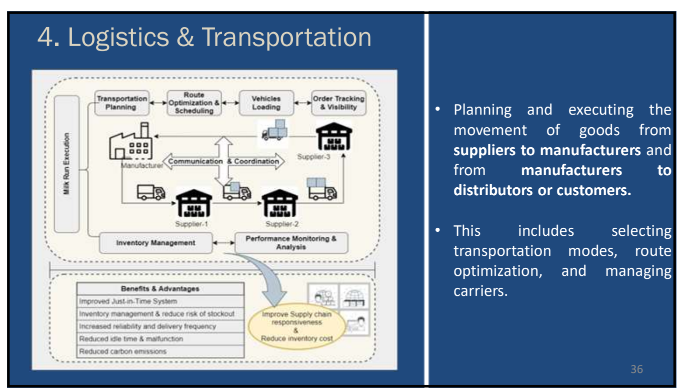
4. Logistics & Transportation
Transportation Planning
Route Optimization & Scheduling
Vehicles Loading
Order Tracking & Visibility
Milk Run Execution
Manufacturer
Communication & Coordination
Supplier-3
Supplier-1
Supplier-2
Inventory Management
Performance Monitoring & Analysis
Benefits & Advantages
Improved Just-in-Time System
Inventory management & reduce risk of stockout
Increased reliability and delivery frequency
Reduced idle time & malfunction
Reduced carbon emissions
Improve Supply chain responsiveness
&
Reduce inventory cost
• Planning and executing the movement of goods from **suppliers to manufacturers** and from **manufacturers to distributors or customers.**
• This includes selecting transportation modes, route optimization, and managing carriers.
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Logistics & Supply Chain Management

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Logistics & Supply Chain Management

	Supply Chain Management	Logistics Management
Scope	SCM involves- **Coordination** and **integration** of all **activities**, from the procurement of **raw materials to the delivery of finished products** to customers.	Deals with- **Movement** and **storage of goods**, materials, and information within an organization or between different organizations
Focus	It focuses on **strategic planning, decision-making**, and **optimizing the entire network of suppliers, manufacturers, distributors,** and **customers.**	It focuses on **specific activities** such as **transportation, warehousing, inventory management, order fulfillment, and distribution**.
Time-Horizon	• Longer-term perspective • Strategic planning and decision-making • Align supply chain goals	• Short-term • Day-to-day operational activities • Concerns itself with immediate tasks

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Logistics & Supply Chain Management

	Supply Chain Management	Logistics Management
Integration	SCM seeks to integrate all functions and activities along the supply chain.	Logistics activities are often more siloed and may function independently within an organization.
Objectives	To achieve higher-level goals, such as • Improving customer satisfaction, • Reducing lead times, • Minimizing inventory costs, • Enhancing overall supply chain resilience, and • Driving competitive advantage	To ensure the **timely and cost-effective movement of goods from one point to another**
Decision Making	• SCM involves strategic decision-making, Example: Supplier selection, Demand forecasting, network design, inventory strategy, and risk management, among others	• Logistics decisions are typically tactical and operational in nature. Example: selecting carriers, optimizing delivery routes, and managing warehouse staffing.

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5. Distribution & Fulfillment

Ensures that customer orders are fulfilled accurately and promptly.

Ensuring the product reach customer in timely and Cost-effective manner

This can involve managing **warehouses, distribution centers,** and **order fulfillment processes.**

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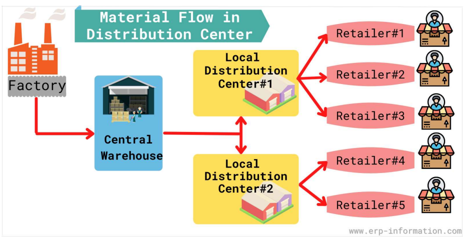
Distribution Resource Planning

Distribution Resource Planning is a systematic approach to managing the distribution phase of the supply chain.



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Distribution Resource Planning (DRP)

- (DRP) is a comprehensive approach to **managing the flow of goods from manufacturers to end customers**.
- It is a subset of supply chain management that **focuses** specifically **on the distribution phase.**
- DRP aims to ensure that the **right products are delivered to the right place at the right time** while minimizing costs and maximizing efficiency.

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Key components and principles of **DRP**

- **Inventory Management:** Efficiently manage inventory levels across distribution network.
- **Demand Forecasting:** Predict future demand to plan production and distribution activities.
- **Order Processing:** Streamline order processing workflow for timely fulfillment.
- **Transportation Management:** Optimize transportation routes and methods for cost-effective delivery.
- **Warehousing and Storage:** Optimize warehouse layouts and processes for efficient storage.
- **Information Systems:** Utilize technology for real-time visibility and data-driven decision making.
- **Collaboration and Coordination:** Foster collaboration among stakeholders for streamlined operations.

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6. Demand Forecasting

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Types of Forecasts by Time Horizon

- Short-range forecast
  - ✓ Usually < 3 months
    - Job scheduling, worker assignments
- Medium-range forecast
  - ✓ 3 months to 2 years
    - Sales/production planning
- Long-range forecast
  - ✓ > 2 years
    - New product planning

Quantitative methods

Detailed use of system

Design of system

Qualitative Methods

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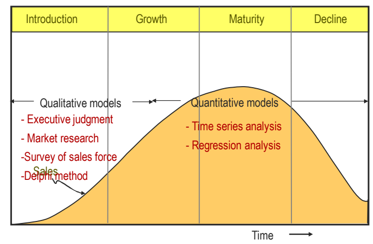
Forecasting During the Life Cycle



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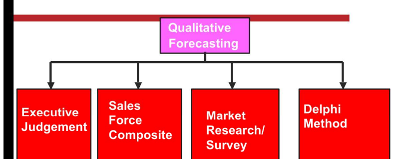
Qualitative Forecasting Methods

Qualitative Forecasting Methods



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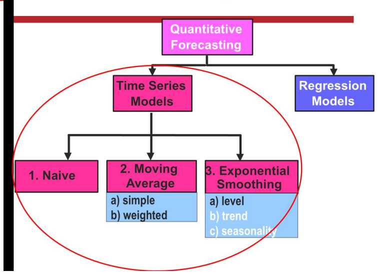
Quantitative Forecasting Methods



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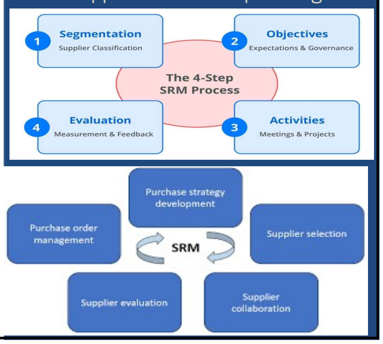
7. Supplier Relationship management

Building and maintaining strong relationship with suppliers

Why??

To ensure a consistent & reliable supply of materials & components



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Supplier Relationship management

SRM is often seen as a strategic part of procurement and supply chain management, focusing on collaboration, communication, and long-term relationships rather than just transactional interactions.

Key Components of SRM-

1. Supplier Segmentation
2. Performance Management
3. Risk Management
4. Collaboration and Communication
5. Contract Management
6. Supplier Development
7. Technology and Data Management

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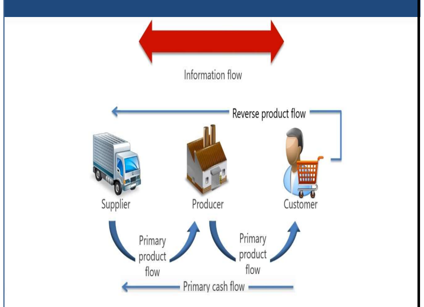
8. Information Flow

Data related to Order
Inventory Level
Demand Forecast
Physical procurement decisions
Manufacturing decisions

Managing the flow of information across the supply chain

It involves the use of technology & information system like ERP and SCM Software



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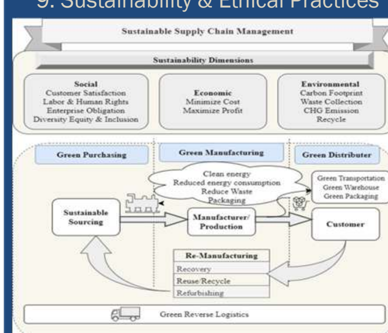
9. Sustainability & Ethical Practices

Supply Chain Management includes consideration of sustainable and ethical practices such as:

1. Responsive Sourcing
2. Reducing Environmental impact
3. Ensuring fair carbon practices



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- Natural Disasters
- Geopolitical Risks
- Epidemics
- Terrorist Attacks
- Environmental Risks
- Volatile Fuel prices
- Rising Labor Costs
- Currency fluctuations
- Port Delays
- Market Changes
- Supplier's Performance
- Forecasting Accuracy
- Execution Problems

10. Risk Management

1. Identifying Risk >> 2. Quantifying & Prioritizing Risk >> 3. Mitigation of Risk >> 4. Speeding Recovery

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Current Issues/Problems in SCM

Coordinating	• Coordinating the relationships between mutually supportive but separate organizations
Raising	• Raising senior management awareness of SCM as a significant competitive weapon.
Managing	• Managing customer touch points
Optimizing	• Optimizing global supplier, production, & distribution networks

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Careers in SCM

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Thank You

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